

MATRIX MINDS

Machine Learning

The complete workflow from raw data to a deployed, monitored model.

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Chapter 1 — The Machine Learning Mindset

Machine Learning is the science of building programs that improve at a task with experience, measured by a performance metric. The classic formulation by Tom Mitchell: a program learns from experience E with respect to task T and performance P , if its performance at T , measured by P , improves with E .

Three families

Supervised (labelled data), Unsupervised (no labels), Reinforcement (reward signal). 90% of business ML is supervised.

Chapter 2 — The End-to-End ML Workflow

1) Frame the problem as a prediction task. 2) Collect and label data. 3) Split into train/validation/test. 4) Engineer features. 5) Train candidate models. 6) Evaluate on validation. 7) Tune hyperparameters. 8) Test once on held-out test set. 9) Deploy. 10) Monitor for drift.

The most common mistake

Touching the test set more than once. Every time you peek and re-tune, you leak information and overestimate real-world performance.

Chapter 3 — Data: The Real Bottleneck

Models are commodities; clean labelled data is the moat. Spend 70% of your time on data quality, deduplication, label consistency, and balanced sampling.

Feature engineering

Convert raw data into informative inputs: one-hot encoding for categoricals, scaling for numerics, date decomposition (day-of-week, hour), text → TF-IDF or embeddings, target encoding for high-cardinality categoricals (with care to avoid leakage).

Handling imbalance

For rare-class problems (fraud, churn), use class weights, oversampling (SMOTE), or threshold tuning — not just accuracy.

Chapter 4 — Algorithms You Will Actually Use

Linear & Logistic Regression

Fast, interpretable, strong baselines. Always start here.

Decision Trees & Random Forests

Handle non-linearities and mixed feature types with almost no preprocessing.

Gradient Boosted Trees

XGBoost, LightGBM, CatBoost — the workhorses of tabular ML and Kaggle competitions.

k-Nearest Neighbours

Simple, no training, but slow at inference and weak in high dimensions.

Support Vector Machines

Strong on small/medium datasets with clear margins; rarely first choice today.

Neural Networks

Best when data is large, unstructured (images, audio, text), or you have plenty of compute.

Chapter 5 — Evaluation Metrics, Done Right

Classification

Accuracy lies on imbalanced data. Prefer precision, recall, F1, ROC-AUC, and PR-AUC. Report a confusion matrix.

Regression

MAE is robust and interpretable, RMSE punishes large errors, R^2 compares against the mean baseline.

Ranking & Recommendations

NDCG, MAP, MRR — order matters, not just whether the right item is present.

Chapter 6 — Cross-Validation and Hyperparameter Tuning

k-fold cross-validation gives a more honest estimate of generalisation. For time series, use forward-chaining splits — never random shuffle. Stratify on the target for classification.

Tuning

GridSearch is exhaustive; RandomSearch is usually as good in a fraction of the time; Bayesian optimisation (Optuna) is best for expensive models.

Chapter 7 — Putting a Model into Production

A model.pkl on your laptop is not a product. Production needs: a versioned model registry, a REST or batch interface, input validation, latency and error monitoring, and a rollback plan.

MLOps essentials

Track experiments (MLflow, Weights & Biases), version data (DVC), automate retraining pipelines (Airflow, Prefect), and monitor for feature drift and concept drift.

Chapter 8 — A 6-Week Self-Study Plan

Week 1: Python, NumPy, Pandas. Week 2: Scikit-learn end-to-end on the Titanic and California Housing datasets. Week 3: Trees and boosting. Week 4: Neural networks with PyTorch. Week 5: Deploy a model behind a FastAPI endpoint. Week 6: Compete in a Kaggle Getting Started competition and publish your notebook.

Closing

Master the workflow before chasing the latest model. — S. Hareedh, Matrix Minds

Thank you for reading.

This eBook was drafted by **S. Hareedh**, Founder of Matrix Minds. If it helped you, share it with one friend who is starting out. For feedback, write to **matrixmindsha@gmail.com**.